

Assignment 2

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Q1. Online Food Delivery Application

food/model/FoodOrder.java

```
package food.model;

public abstract class FoodOrder { private int orderId;
private String customerName; private double amount;
private static String restaurantName = "Tasty Bites"; private static int orderCount = 0;

public FoodOrder(int orderId, String customerName, double amount) { this.orderId = orderId;
this.customerName = customerName; this.amount = amount; orderCount++;
}

public int getOrderId() { return orderId; }
public void setOrderId(int orderId) { this.orderId = orderId; }

public String getCustomerName() { return customerName; }
public void setCustomerName(String customerName) { this.customerName = customerName; }

public double getAmount() { return amount; }
public void setAmount(double amount) { this.amount = amount; } public static String getRestaurantName() { return restaurantName; }
public abstract double calculateDeliveryCharge();
public static void displayTotalOrders() { System.out.println("Total Orders: " + orderCount);
}
}
```

food/model/Discountable.java

```
package food.model;

public interface Discountable { double applyDiscount();
}
```

food/model/RegularOrder.java

```
package food.model;
public class RegularOrder extends FoodOrder implements Discountable
{ public RegularOrder(int orderId, String customerName, double amount) {
super(orderId, customerName, amount);
}

@Override
public double calculateDeliveryCharge() { return 80.0;
}

@Override
public double applyDiscount() { return getAmount() * 0.10;
}
}
```

food/model/PremiumOrder.java

```

package food.model;
public class PremiumOrder extends FoodOrder implements Discountable
{ public PremiumOrder(int orderId, String customerName, double amount) {
super(orderId, customerName, amount);
}

@Override
public double calculateDeliveryCharge() { return 50.0;
}

@Override
public double applyDiscount() { return getAmount() * 0.15;
}
}

```

food/utility/OrderUtility.java

```

package food.utility; public class OrderUtility {
public static boolean validateAmount(double amount) { return amount > 0;
}

public static boolean validateCustomerName(String name) { return name != null && !name.trim().isEmpty();
}

public static String generateOrderSummary(int orderId, String customerName, double amount,
double discount, double deliveryCharge, double finalAmount) { return "OrderID: " + orderId + ", Customer: " + customerName +
", Bill: " + amount + ", Discount: " + discount +
", Delivery: " + deliveryCharge + ", Final: " + finalAmount;
}
}

```

food/main/MainApp.java

```

package food.main; import food.model.*;
import food.utility.OrderUtility;

public class MainApp {
public static void main(String[] args) { FoodOrder[] orders = new FoodOrder[6]; orders[0] =
new RegularOrder(1, "Aman", 500); orders[1] = new PremiumOrder(2, "Riya", 800); orders[2] =
new RegularOrder(3, "Karan", 350);
orders[3] = new PremiumOrder(4, "Simran", 1200); orders[4] = new RegularOrder(5, "Vikram", 600); orders[5] =
new PremiumOrder(6, "Neha", 950);

for (FoodOrder order : orders) {
double deliveryCharge = order.calculateDeliveryCharge(); double discount = ((Discountable) order).applyDiscount();
double finalAmount = order.getAmount() - discount + deliveryCharge; if (OrderUtility.validateAmount(order.getAmount()) &&
OrderUtility.validateCustomerName(order.getCustomerName()))
{ System.out.println(OrderUtility.generateOrderSummary(order.getOrderId(),
order.getCustomerName(), order.getAmount(), discount, deliveryCharge, finalAmount));
}
}
FoodOrder.displayTotalOrders();
}
}

```

Q2. Smart Library Resource Management System

library/model/LibraryResource.java

```
package library.model;

public abstract class LibraryResource { private int resourceId;
private String title; private String author;
private static String libraryName = "Central Library"; private static int resourceCount = 0;

public LibraryResource(int resourceId, String title, String author) { this.resourceId = resourceId;
this.title = title; this.author = author; resourceCount++;
}

public int getResourceId() { return resourceId; }
protected void setResourceId(int resourceId) { this.resourceId = resourceId; }

public String getTitle() { return title; } public String getAuthor() { return author; }

public static String getLibraryName() { return libraryName; } public abstract double calculateFine(int overdueDays);
public static void displayTotalResources() { System.out.println("Total Resources: " + resourceCount);
}
}
```

library/model/Printable.java

```
package library.model; public interface Printable {
void printDetails();
}
```

library/model/Book.java

```
package library.model;
public class Book extends LibraryResource implements Printable { public Book(int resourceId, String title, String author) {
super(resourceId, title, author);
}

@Override
public double calculateFine(int overdueDays) { return overdueDays * 5.0;
}

@Override
public void printDetails() {
System.out.println("Book -> ID: " + getResourceId() +
", Title: " + getTitle() + ", Author: " + getAuthor());
}
}
```

library/model/DigitalResource.java

```
package library.model;
public class DigitalResource extends LibraryResource implements Printable {

public DigitalResource(int resourceId, String title, String author) { super(resourceId, title, author);
}

@Override
public double calculateFine(int overdueDays) { return overdueDays * 2.0;
}

@Override
public void printDetails() {
System.out.println("DigitalResource -> ID: " + getResourceId() + ", Title: " + getTitle() + ", Author: " + getAuthor());
}
}
```

library/util/InputValidator.java

```

package library.util;

public class InputValidator {

    public static boolean validateResourceId(int id) { return id > 0;
}

    public static boolean validateFineDays(int days) { return days >= 0;
}
}

```

library/main/MainApp.java

```

package library.main; import library.model.*;
import library.util.InputValidator;

public class MainApp {
    public static void main(String[] args) { LibraryResource[] resources = new LibraryResource[5];
    resources[0] = new Book(1, "Java Programming", "James Gosling");
    resources[1] = new DigitalResource(2, "Cloud Computing Basics", "Amit Sharma");
    resources[2] = new Book(3, "Data Structures", "Robert Lafore");
    resources[3] = new DigitalResource(4, "AI Fundamentals", "Andrew Ng");
    resources[4] = new Book(5, "Operating Systems", "Galvin");

    int[] overdueDays = {3, 5, 0, 7, 2}; double totalFine = 0;

    for (int i = 0; i < resources.length; i++) { LibraryResource resource = resources[i];
    if (InputValidator.validateResourceId(resource.getResourceId()) && InputValidator.validateFineDays(overdueDays[i]))
    { ((Printable) resource).printDetails();
    double fine = resource.calculateFine(overdueDays[i]); System.out.println("Overdue Days: " + overdueDays[i] + ", Fine: Rs. " + fine);
    totalFine += fine;
    }
    System.out.println("Total Fine of all overdue resources: Rs. " + totalFine); LibraryResource.displayTotalResources();
}
}

```